Lab 09 – (Week of October 21st)

All of the activities of this lab are to be completed in teams of two (or three, but only as necessary), and each team is limited to the use of one computer. The activities involving use of the "Practice-It" website are to be done through just one of the student authorizations.

Activity 1: Practice-It Self-Checks and Exercises

After logging in to Practice-It, click on the link labeled **Building Java Programs, 4th Edition** and then on the link indicating **BJP Chapter 4**.

Do the following Self-Checks and Exercises and, when finished, show your GTA (graduate teaching assistant) that you've completed all of them correctly. (Go back to the web page that lists all the Chapter 4 Self-Checks and Exercises and refresh it. Any that were answered correctly will be indicated on the page.)

1. **BJP4 Self-Check 4.5 (ifElseMystery1):** Show output.
2. **BJP4 Self-Check 4.8 (ScannerLogicError):** Correct bug. Note that the code fragment is intended to print "`Odd`" when the user inputs a number not evenly divisible by two, and print "`Divisible by 6`" when the input is evenly divisible by six.
3. **BJP4 Self-Check 4.9 (favoriteColor):** Correct a bug.
4. **BJP4 Self-Check 4.10 (javaIsAwesome):** Refactor the code.
5. **BJP4 Self-Check 4.11 (contribution):** Refactor code.
6. **BJP4 Self-Check 4.15 (sumTo):** Correct the error.
7. **BJP4 Self-Check 4.16 (countFactors**). Correct the error.
8. **BJP4 Self-Check 4.27 (medianOf3):** Correct the error in a method.
9. **BJP4 Exercise 4.19 (quadrant**).

Activity 2: `quadrant` method and Driver

This activity is related to **BJP4 Exercise 4.19 (quadrant**). After correcting the method you are to then incorporate it into the given `quadrantDriver.java` source program, and test it thoroughly.

Once you are convinced that you have this method working fully correctly you are to show it to your instructor and demonstrate it using the `quadrantDriver` program. Only then submit your completed `quadrantDriver` program to the CMPS 134L CWS for the section you are working in.

Activity 3: Practice-It Self-Checks and Exercises

After logging in to Practice-It, click on the link labeled **Building Java Programs, 4th Edition** and then on the link indicating **BJP Chapter 5**.

Do the following Self-Checks and Exercises and, when finished, show your GTA (graduate teaching assistant) that you've completed all of them correctly. (Go back to the web page that lists all the Chapter 5 Self-Checks and Exercises and refresh it. Any that were answered correctly will be indicated on the page.)

1. **BJP4 Self-Check 5.1 (whileLoops):** How many iterations?
2. **BJP4 Self-Check 5.3 (whileLoopMystery1):** Show output.
3. **BJP4 Self-Check 5.8 (doWhileLoops)**: How many iterations?
4. **BJP4 Self-Check 5.14 (logicExpressions2)**: Determine results of `boolean` expressions.

Activity #4: `isPalindrome` method and the `isPalindromeDriver.java` Program

This activity is similar to **BJP4 Exercise 4.16 (printPalindrome**), except that instead of printing, the method is to be a functional method, named `isPalindrome`, that returns either `true` or `false` indicating whether its argument is a "palindrome".

A palindrome is a string that is "spelled" the same way forwards as it is backwards. Examples include "abba", "racecar" and "mom". In order to improve your thinking regarding the use of `while` and `do` loops, you are **not** to use a `for` loop in your implementation of this method!

You are to provide your implementation for this method in the given program, `isPalindromeDriver.java`, and use it to develop, test and then demonstrate your solution to the Lab Instructor.Consider the dialogue shown to the right.

```
c:\C134L>java isPalindromeDriver
Enter a String:>Hello
"Hello" is not a palindrome!
Enter a String:>mom
"mom" is a palindrome!
Enter a String:>w00w
"w00w" is a palindrome!
Enter a String:>wooow
"wooow" is a palindrome!
Enter a String:>w0x0w
"w0x0w" is a palindrome!
Enter a String:>racecar
"racecar" is a palindrome!
Enter a String:>X
"X" is a palindrome!
Enter a String:>
Done!!!
```

Once you are convinced that you have the `isPalindrome` method working you are to show it to your instructor and demonstrate it using the `isPalindromeDriver` program. Only then submit the file to the CMPS 134L CWS for the section you are working in.

Activity #5: `wordCount` method and the `wordCountDriver.java` Program

This activity is basically **BJP4 Exercise 4.18 (wordCount)**. You are to write a functional method, named `wordCount`, that accepts a `String` as its parameter and returns the count of the number of words in the given parameter. For the purposes of this question, a word is a sequence of one or more non-space characters (any character other than ' ').

For example:
`wordCount("hello")` returns 1
`wordCount("how are you?")` returns 3
`wordCount("  this     string  ")` returns 2
`wordCount(" ")` and `wordCount("")` returns 0.

You are to provide your implementation for this method in the given program, `wordCountDriver.java`, and use it to develop, test and then demonstrate your solution to the Lab Instructor. Consider the sample execution shown to the right.

```
c:\C134L\Lab08_Oct15\Solution>java wordCountDriver DoI.txt

"Declaration of Independence Excerpt" contains 4 words!
"We hold these truths to be self-evident, that" contains 8 words!
"all men are created equal, that they are endowed by" contains 10 words!
"their Creator with certain unalienable Rights, that" contains 7 words!
"among these are Life, Liberty and the pursuit of" contains 9 words!
"Happiness. That to secure these rights, Governments" contains 7 words!
"are instituted among Men, deriving their just powers" contains 8 words!
"from the consent of the governed. That whenever" contains 8 words!
"any Form of Government becomes destructive of these" contains 8 words!
"ends, it is the Right of the People to alter or to" contains 12 words!
"abolish it, and to institute new Government, laying" contains 8 words!
"its foundation on such principles and organizing its" contains 8 words!
"powers in such form, as to them shall seem most" contains 10 words!
"likely to effect their Safety and Happiness." contains 7 words!
"Prudence, indeed, will dictate that Governments long" contains 7 words!
"established should not be changed for light and" contains 8 words!
"transient causes; and accordingly all experience" contains 6 words!
"hath shewn, that mankind are more disposed to suffer," contains 9 words!
"while evils are sufferable, than to right themselves" contains 8 words!
"by abolishing the forms to which they are" contains 8 words!
"accustomed." contains 1 words!
"" contains 0 words!
Done!!!
```

Note that this given driver program reads its input from a file (in this case the given file, `DoI.txt`). Thus, it will be necessary for you to configure jGRASP to present the "Run Arguments" text area and to type the name of the input file in that area.

Once you are convinced that you have the `wordCount` method working you are to show it to your instructor and demonstrate it using the `wordCountDriver` program. Only then submit the file to the CMPS 134L CWS for the section you are working in.